

# Client screens — every end-user surface, in depth (light + dark, phone→tablet→desktop→TV)

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Master technical specification — Volume 10 (Design System), nano-detail deep-dive. This document **owns** the complete **screen-level UX specification of the HelixVPN Client app** (flavor access / "Helix Access" — the end-user VPN application). It specifies **every screen** the end user can reach: purpose, region layout, the **responsive behaviour across all four form factors** (phone / tablet / desktop / TV-leanback), the components each composes (cited to the sibling [component-library.md]), the **states** each renders (loading / empty / error / success + the **7 connection states** where relevant), in/out navigation, key interactions, accessibility, **light + dark** notes, and an **ASCII wireframe** (plus a Mermaid navigation map). It is **original HelixVPN UX design work**.

**The product's emotional centre of gravity** is one question — "*am I protected right now?*" [04\_CLIENT §7]. Every screen is designed so that question is answered instantly and unambiguously, and so the **loading/connecting** state is never mistaken for the **connected** state (§11.4.107 — a spinner is **not** proof of connection; the Home screen distinguishes them with colour **and** icon **and** text **and** an independent "tunnel-up" signal).

**SPEC-ONLY.** It describes *what each screen is and how it lays out and behaves* — not the shipping Flutter widget tree. The component internals are owned by [component-library.md]; the colours by

[color-system.md]; the token structure/scales by [design-tokens.md]; the type/icon/motion by [typography-iconography-motion.md]; the Console screens by [screens-console.md]; the Connector-appliance screens by [screens-connector.md]. This doc consumes all of those.

**Boundary with sibling docs.** Owns: the Client **screen inventory**, each screen's regions / responsive rules / state matrix / nav / ally / wireframe. Consumes: the **7-variant ffi::TunnelStatus** the Home screen switches on [FFI §3.2]; the connection-state palette + contrast proofs [COLOR §3–§4]; the token scales (spacing/breakpoints/radius/motion) [DT §6]; the Client capability set {tunnel, account, splitTunnel} and the AdaptiveScaffold responsive contract [04\_CLIENT §6, §7.3]; the named signature components (ConnectButton, StatusChip, ExitPicker, ShieldIndicator, NetworkTile) [04\_CLIENT §7.2].

**Evidence base.** [04\_CLIENT §N] = final/03-client-core-and-ui.md; [FFI §N] = final/v04-client/ffi-surface.md; [UI §N] = final/v04-client/helix-ui-flutter.md; [COLOR §N] = final/v10-design/color-system.md; [DT §N] = final/v10-design/design-tokens.md; [SPINE §N] = final/SPECIFICATION.md. Claims not grounded in the evidence base or this document's own original UX design choices are tagged UNVERIFIED per constitution §11.4.6 — never fabricated. [component-library.md] is a **same-wave sibling** that this doc forward-references for every component; component internals are owned there.

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## Table of contents

- 0. Scope, platforms & the §11.4.162 layout covenant
- 1. Screen inventory (table)
- 2. Navigation map
- 3. The responsive shell — AdaptiveScaffold
- 4. Onboarding / first-run
  - 4.1 Welcome
  - 4.2 Sign-in (OIDC vs anonymous device token)
  - 4.3 VPN permission grant
- 5. Connect / Home (the hero)
- 6. Exit selection
- 7. Multi-hop builder
- 8. Shields / Privacy settings
- 9. Account
- 10. Settings
- 11. Error / Danger states
- 12. Connector-mode toggle

- [13. Cross-screen patterns \(loading/empty/error, ally, motion\)](#)
- [14. Surfaced decisions & cross-doc contracts](#)
- [Sources verified](#)

## 0. Scope, platforms & the §11.4.162 layout covenant

This document specifies the **Client app** (`HelixFlavor.access`, "Helix Access") — the consumer VPN application whose home is the big `ConnectButton` [04\_CLIENT §6]. Its capability set is `{tunnel, account, splitTunnel}`; it links the real Rust `helix-core` via FFI and a per-platform `TunnelPlatform` shim [04\_CLIENT §6, §4]. (The **Console** admin app and the **Connector** appliance app have their own screen docs; this doc cross-references them only at the boundaries — §9, §12.)

**Form factors.** Every screen is specified across **four** size classes, branched on **width, never on Platform.isX** [04\_CLIENT §7.3] so a resized desktop window behaves like a phone and Web/desktop "just work":

| Size class      | Width (token [DT §6.6])                                        | Shell                                                                | Primary device archetype        |
|-----------------|----------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------|
| <b>compact</b>  | < <code>breakpoint.md</code> (600 px)                          | <code>BottomNavigationBar</code> , single pane                       | phone (portrait)                |
| <b>medium</b>   | 600–1024 px                                                    | <code>NavigationRail</code> , master/detail                          | large phone landscape / tablet  |
| <b>expanded</b> | > <code>breakpoint.lg</code> (1024 px)                         | extended <code>NavigationRail</code> , multi-pane                    | desktop (macOS/Windows/Linux)   |
| <b>tv</b>       | ≥ <code>breakpoint.xl</code> (1440 px) + <b>leanback input</b> | extended rail, <b>enlarged focus targets</b> , D-pad focus traversal | Android TV / leanback HarmonyOS |

**TV-leanback is a size class + an input model**, not a separate platform tree. It renders the expanded layout with: focus targets ≥ `space.scale.7` (48 px → enlarged to ~64 px hit area), a **visible focus ring** (`border.focus`, [COLOR §2.3]) on every focusable, D-pad/remote traversal order declared per screen, and no hover-only affordances. UNVERIFIED — the exact TV focus-order matrices per screen are

pinned by the platform-adaptation spec [platform-adaptation.md] + the leanback golden-screenshot suite; this doc states the focus *intent* per screen, frozen there.

**Platforms the Client ships on (7 of the 8).** iOS, macOS, Android, Windows, Linux, HarmonyOS NEXT, Aurora OS. **Web is NOT a Client target** — the Web build is the Console only (it lacks Capability.tunnel, so it never links the Rust staticlib) [04\_CLIENT §5.7, §6]. Every screen below therefore assumes a native host with a TunnelPlatform shim available.

**The §11.4.162 layout covenant (applies to every screen below).** Per the OpenDesign mandate: **no element overlaps another; no element overlays a label; every screen ships a light AND a dark variant; every screen is covered by visual regression** [visual-regression-and-qa.md]. At the layout level this doc enforces:

- 1. **One column of meaning at a time on compact.** Regions stack vertically with a space.scale.4 (16 px) gap minimum; nothing floats over body text except the intentional overlay surfaces (sheets, the Danger banner) which always sit on a scrim ([COLOR §5.2]).
- 2. **The connection state owns the screen** [04\_CLIENT §7.2]; secondary chrome recedes. Only **one** saturated colour (the active state) is on screen at a time ([COLOR §0.2]).
- 3. **Labels are never drawn on a coloured fill that is not contrast-proven** ([COLOR §4.3]); state cross-fades swap fill **and** label together (motion.semantic.stateXfade, 180 ms [DT §6.4]) so no in-between frame is illegible.
- 4. **The Danger banner is z-top** (z.semantic.danger 2000 [DT §6.5]) and can never be occluded — even by a modal or a toast.

## 1. Screen inventory (table)

Routes are go\_router-style paths (UNVERIFIED exact router config — owned by [navigation-and-routing.md], a planned sibling; the *paths* + *hierarchy* here are this doc's design intent). "Key components" cite [component-library.md].

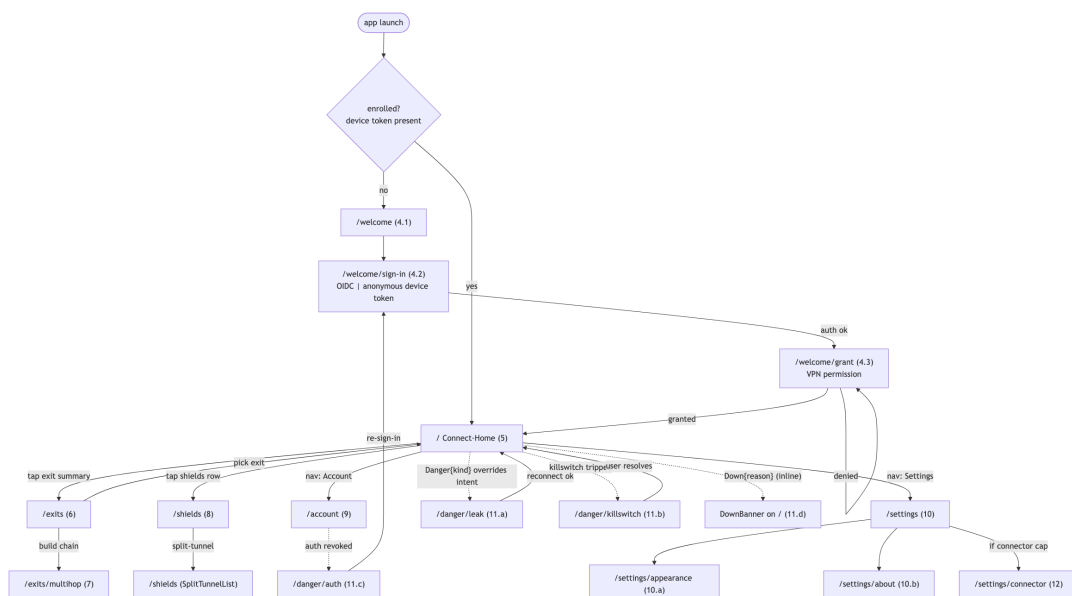
| #   | Screen  | Route            | Key components                                            | States rendered                      |
|-----|---------|------------------|-----------------------------------------------------------|--------------------------------------|
| 4.1 | Welcome | /welcome         | OnboardPager, HeroIllustration, PrimaryButton, TextButton | static (no async)                    |
| 4.2 | Sign-in | /welcome/sign-in | AuthChoiceCard ×2, OidcWebView/                           | idle · in-progress · error · success |

| #    | Screen                | Route                | Key components                                                                                       | States rendered                                  |
|------|-----------------------|----------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 4.3  | VPN permission grant  | /welcome/grant       | DeviceCodePanel, InlineError<br><br>PermissionExplainer, PrimaryButton, PlatformConsentSheet (OS)    | prompting · granted · denied · re-prompt         |
| 5    | <b>Connect / Home</b> | / (home)             | <b>ConnectButton</b> , <b>StatusChip</b> , ExitSummaryCard, <b>ShieldIndicator</b> row, DangerBanner | <b>all 7 TunnelStatus</b> + loading-vs-connected |
| 6    | Exit selection        | /exits               | <b>ExitPicker</b> , SearchField, FavoriteToggle, ExitRow (rtt + jurisdiction), SegmentTabs           | loading · empty · error · populated · selected   |
| 7    | Multi-hop builder     | /exits/multihop      | HopChainEditor, HopRow ×N, AddHopButton, ChainSummaryBar                                             | empty(1 hop) · building · invalid · valid        |
| 8    | Shields / Privacy     | /shields             | <b>ShieldIndicator</b> tiles, SwitchTile, SplitTunnelList, CostNoteSheet                             | each shield: on · off · unavailable · pending    |
| 9    | Account               | /account             | IdentityCard, SubscriptionCard*, DeviceList/ DeviceRow, SignOutButton                                | loading · loaded · offline-cached · error        |
| 10   | Settings              | /settings            | SettingsGroup, ThemeSelector, NotificationToggles, AboutPanel                                        | static + per-toggle async                        |
| 10.a | Appearance            | /settings/appearance | ThemeSelector (light/dark/system), AccentPreview                                                     | applied immediately                              |
| 10.b | About                 | /settings/about      | AboutPanel, VersionRow, LicensesList, DiagnosticsButton                                              | static                                           |
| 11.a |                       |                      |                                                                                                      | Danger{kind:"leak"}                              |

| #    | Screen                     | Route                       | Key components                                          | States rendered                             |
|------|----------------------------|-----------------------------|---------------------------------------------------------|---------------------------------------------|
|      | Leak detected (full)       | /danger/leak (modal route)  | DangerScreen, DangerBanner, LeakDetail, ReconnectButton |                                             |
| 11.b | Kill-switch tripped (full) | /danger/killswitch          | DangerScreen, KillSwitchDetail, PrimaryButton           | Danger{kind:"killswitch_tripped"}           |
| 11.c | Auth revoked (full)        | /danger/auth                | AuthRevokedScreen, ReSignInButton                       | auth-revoked (→ re-enroll)                  |
| 11.d | Connection dropped         | inline banner on /          | DownBanner                                              | Down{reason}                                |
| 12   | Connector-mode toggle      | /settings/connector (gated) | ConnectorModeCard, AdvertiseCidrList, SwitchTile        | unavailable* · off · advertising · conflict |

\* SubscriptionCard and the connector capability are **flavor/deployment conditional** — see §9 (subscription is UNVERIFIED for the self-hostable MVP) and §12 (connector-in-client is a surfaced decision D-SC-3).

## 2. Navigation map

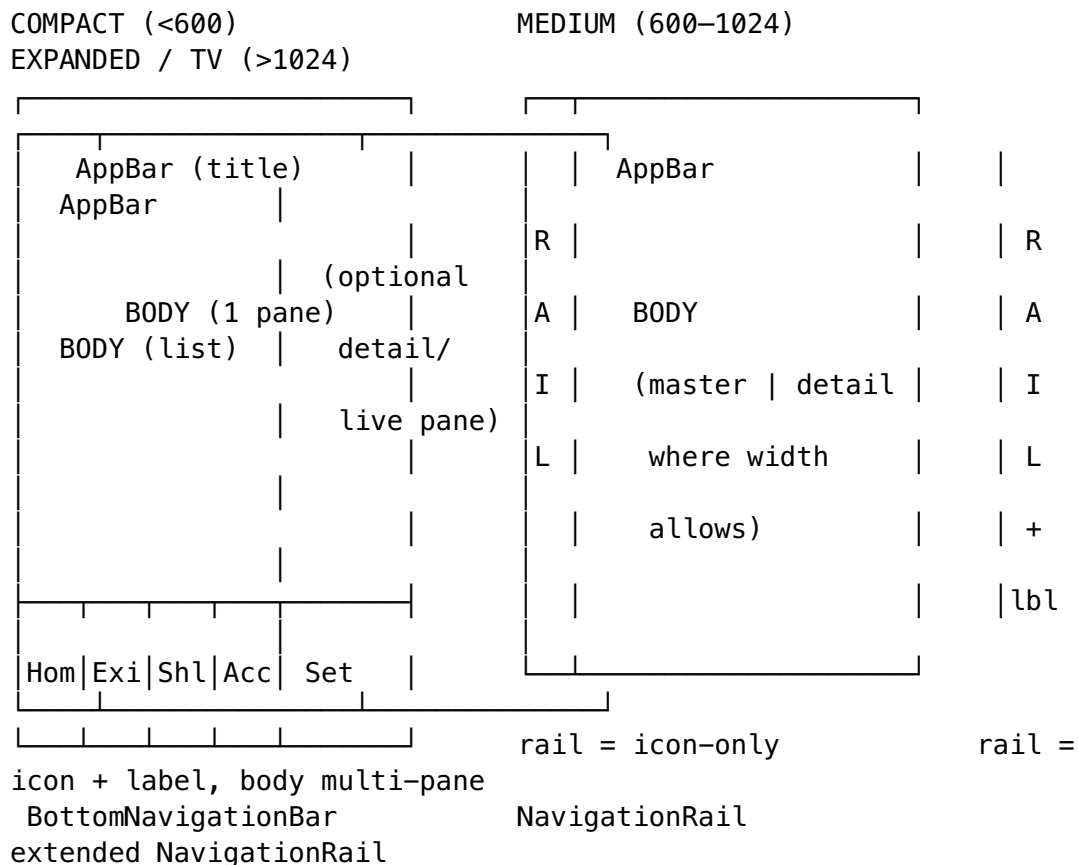


- **First-run gate.** Launch checks for a persisted device token / session [04\_CLIENT §8]. Absent → onboarding (§4); present → straight to Home.

- **Home is the root** of the post-onboarding `AdaptiveScaffold`; Exits, Shields, Account, Settings are its peer destinations (bottom-nav on compact, rail on medium+).
- **Danger routes are pushed over everything** and are **input-overriding** — a `Danger{kind}` from the status stream forces the Danger surface regardless of where the user is ([FFI §3.2], "Danger overrides any user intent"). Down is an **inline banner** on Home (a drop, retrying — not an exposure), distinct from Danger (red, full surface).

### 3. The responsive shell — `AdaptiveScaffold`

Every post-onboarding screen renders inside one shell whose navigation affordance is chosen by **size class** [04\_CLIENT §7.3]. The shell is a single widget; the size class flips its nav region and pane count — the body screens are identical across classes.



- **compact:** `BottomNavigationBar`, 5 destinations (Home · Exits · Shields · Account · Settings), single body pane, full-width `ConnectButton` on Home.
- **medium:** `NavigationRail` (icon-only), body becomes master/detail where the screen defines one (Exits list | selected-exit detail; Settings group | sub-page).
- **expanded:** extended `NavigationRail` (icon + label), multi-pane bodies.

- **tv:** the expanded layout, focus-traversal model (§0), enlarged targets, no bottom nav (D-pad reaches the rail).

A11y: the active destination carries selected semantics; the rail/bottom-nav is keyboard-traversable on desktop/web-less native; the focus ring is the border. focus token ( $\geq 3.0$  contrast vs surface, both themes [COLOR §4.5]).

Light/dark: shell surface.default (#FFFFFF / #151920), rail/appbar surface.default, selected destination tinted surface.brandTint (helix.50 / helix.900); both proven AAA-legible [COLOR §2.1].

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## 4. Onboarding / first-run

Three steps, each its own route so the OS back gesture is sane and a partial enrollment resumes at the right step. A linear OnboardPager on compact; a single centred card column on medium+; never more than one decision per screen (§0 covenant).

### 4.1 Welcome

**Purpose.** Establish trust + the one value promise ("your traffic, private — your servers or ours") and route into sign-in. No account interaction yet.

**Layout (regions, top→bottom):** hero illustration → headline → 1-line subhead → PrimaryButton "Get started" → TextButton "I already have an account / device". On a multi-slide value carousel, a PageIndicator sits between subhead and the buttons (never overlapping them — space.scale.5 gap).

**Components** [component-library.md]: OnboardPager, HeroIllustration, PrimaryButton, TextButton, PageIndicator.

**States.** Static only (no async). No empty/error/loading.

**Responsive.**

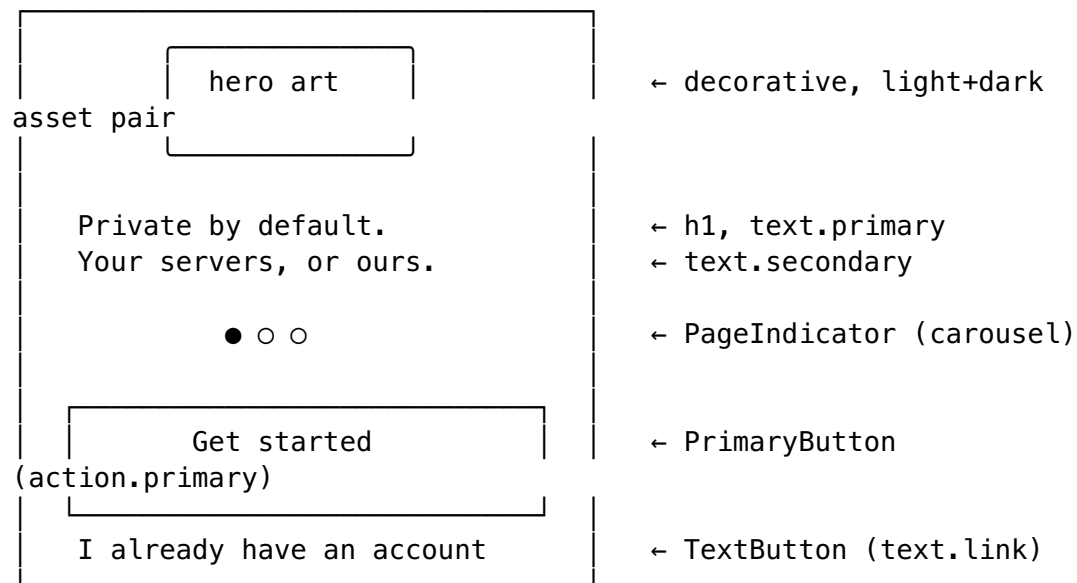
- *phone (compact):* full-bleed hero top ~40 % height, content column 16 px margins (space.scale.4), buttons full-width pinned to safe-area bottom.
- *tablet (medium):* hero + copy in a centred 420-px card, buttons inline width.
- *desktop (expanded):* two-column — illustration left 50 %, copy/actions right, max content width 480 px.
- *TV:* centred, enlarged type (display scale), "Get started" is the default focused control; D-pad down reaches the secondary TextButton.

**Navigation.** in: app launch (unenrolled). out: "Get started" → /welcome/sign-in.

**Interactions.** swipe/arrow between carousel slides; "Get started" advances.

**A11y.** hero illustration is decorative (empty semantic label, not announced); headline is an h1 heading node; buttons carry role + label; carousel exposes "slide N of M" live region; full keyboard/remote traversal.

**Light + dark.** surface.default bg; hero illustration ships **two assets** (light/dark variants per §11.4.162); headline text.primary, subhead text.secondary; primary button action.primary fill + white label.



## 4.2 Sign-in (OIDC vs anonymous device token)

**Purpose.** Enrol the device by one of two paths the spine supports [04\_CLIENT §6 account capability]: **OIDC** (hosted/managed identity) **or** **anonymous device-token** (self-hostable, privacy-max — no personal identity, just a per-device credential). The user picks; both end in a usable session.

**Layout.** Two stacked AuthChoiceCards (compact) / side-by-side (medium+):

- **Card A — Sign in with your provider (OIDC).** Tapping opens an OidcWebView/system browser to the configured issuer; on success the redirect returns a code the client exchanges for a session.
- **Card B — Use this device anonymously (device token).** Tapping shows a DeviceCodePanel: the client generates/registers a device token against the configured server (the server URL is entered here on first run if not preset). No email, no personal identity.

Below the cards: a one-line "Self-hosting? Enter your server" link expanding a `ServerUrlField`.

**Components.** `AuthChoiceCard` ×2, `OIDCWebView` (or system-browser handoff), `DeviceCodePanel`, `ServerUrlField`, `InlineError`, `ProgressIndicator`.

### States.

- *idle*: two choices presented.
- *in-progress*: the chosen path shows a determinate-where-possible `ProgressIndicator` ("Contacting your server..."); the **other** card disables (no double-enrol).
- *error*: `InlineError` under the active card with a typed, human message (unreachable server, bad URL, issuer rejected, token already bound elsewhere); the card returns to idle for retry. Never a silent fail (§11.4.6).
- *success*: advances to `/welcome/grant`.

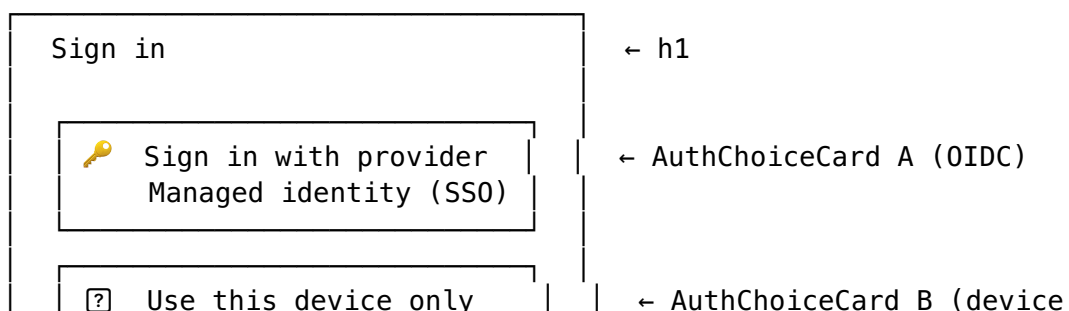
**Responsive.** compact: cards stack, full-width; medium/expanded: cards side-by-side in a centred 640-px row; TV: cards as large focusable tiles, OIDC path uses a device-code/QR flow if no embedded browser (UNVERIFIED — the TV OIDC device-flow is owned by the OIDC integration spec; flagged there).

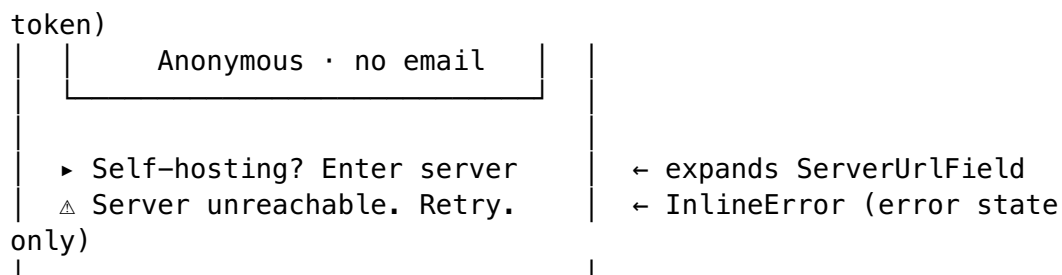
**Navigation.** in: `/welcome` "Get started", or `/danger/auth` re-sign-in (§11.c). out (success): `/welcome/grant` (or directly Home if permission already granted).

**Interactions.** pick a card → run that flow; expand server-URL field; paste/scan server URL; cancel returns to idle.

**A11y.** each card is a single labelled button announcing its path + privacy implication ("anonymous, no personal identity"); the in-progress state announces via live region; errors are announced and focus moves to the `InlineError`.

**Light + dark.** cards surface. `raised`; selected/active card border border. `focus`; OIDC web-view inherits issuer styling (out of our control — flagged in About as third-party); error text feedback. `error` (proven AA both themes [COLOR §2.4]).





## 4.3 VPN permission grant

**Purpose.** Obtain the **OS-level VPN configuration consent** the TunnelPlatform shim requires before the first tunnel can be created [04\_CLIENT §4, §5]. This is a **platform-mediated** step — our screen explains *why*, then triggers the OS consent dialog the platform owns.

**Per-platform OS mechanism** (we explain, the OS prompts) [04\_CLIENT §5]:

| Platform       | OS consent the shim triggers                                                                  |
|----------------|-----------------------------------------------------------------------------------------------|
| iOS / macOS    | NETunnelProviderManager.saveToPreferences → system "... Would Like to Add VPN Configurations" |
| Android        | VpnService.prepare() → system VPN-consent dialog                                              |
| Windows        | install/authorise the privileged tunnel service (UAC elevation)                               |
| Linux          | privileged helper / capability grant for the kernel-WG / tun device                           |
| HarmonyOS NEXT | Network Kit VPN ability authorisation                                                         |
| Aurora OS      | platform tun authorisation                                                                    |

UNVERIFIED — the exact OS dialog copy + the per-platform privileged-service install flow are owned by [platform-adaptation.md] + the per-platform shim specs; this screen owns only the **pre-consent explainer** and the **post-consent state handling**.

**Layout.** PermissionExplainer (icon + 2-line "Helix needs permission to create a VPN tunnel on this device. We never see your traffic.") → PrimaryButton "Allow VPN" → on tap, the OS PlatformConsentSheet appears (OS-rendered) → result handled.

**States.**

- *prompting*: explainer + button (idle).
- *granted*: brief success tick → navigate Home.

- *denied*: a calm re-explain ("Helix can't connect without VPN permission") + "Try again" (re-triggers) + a "How to enable in Settings" deep-link helper. Never dead-ends.
- *re-prompt*: if the OS won't re-show the dialog (already-denied-permanently), we show the Settings deep-link path.

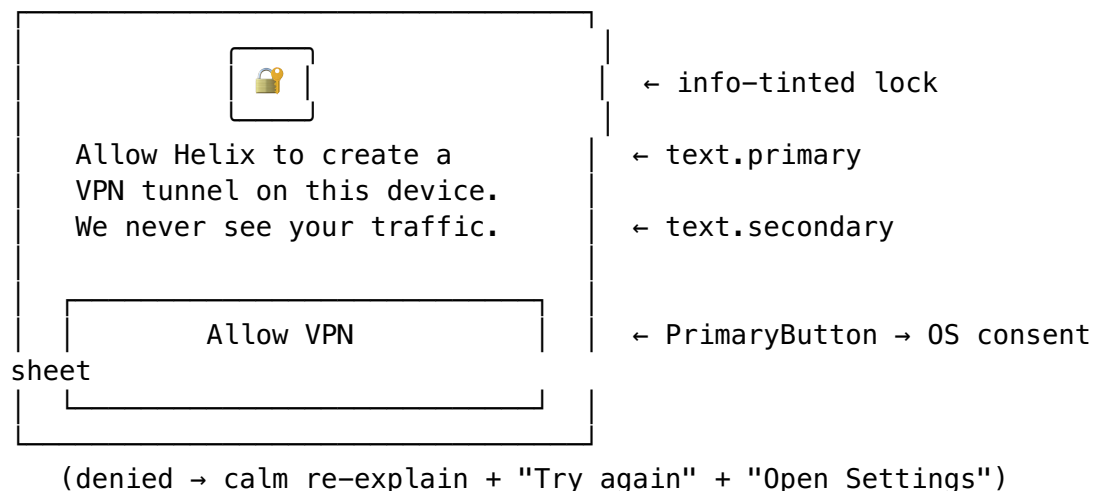
**Responsive.** centred single-column all classes; on desktop the privileged-service install may show an elevation prompt (OS); TV: button default-focused.

**Navigation.** in: /welcome/sign-in success. out (granted): Home. (denied stays.)

**Interactions.** "Allow VPN" → OS sheet; "Try again"; "Open system settings".

**A11y.** explainer is body text + an info icon (decorative); the result state is announced; focus lands on the actionable button in each state.

**Light + dark.** surface.default; explainer icon uses feedback.info; success tick feedback.success; denied re-explain uses feedback.warning (not error — it's recoverable).



## 5. Connect / Home (the hero)

**Purpose.** The product. Answer "*am I protected right now?*" at a glance, let the user connect/disconnect with one tap, and surface the current exit, transport/path, and shields summary. This is the **only** screen that renders **all 7 TunnelStatus variants** [FFI §3.2], and the screen where the **loading/connecting vs connected distinction** is load-bearing (§11.4.107).

**Layout (regions, top→bottom; compact).**

1. AppBar — app title + a small avatar/menu (→ Account).

2. **DangerBanner slot** (hidden unless Danger/Down; z-top when shown).
3. **ConnectButton** — the giant circular tap target, vertically centred, ~space.scale.7-class diameter. Its **fill colour, ring, icon, and label all switch on the status** ([COLOR §3], comp.connectButton.bg.\* [DT §8]).
4. **StatusChip** directly under the button — transport · path · rtt (e.g. MASQUE · direct · 23 ms) when Connected; the in-flight phase text ("Handshaking...") otherwise.
5. **ExitSummaryCard** — current exit name + flag/jurisdiction; tap → /exits.
6. **ShieldIndicator row** — kill-switch / DNS / DAITA / PQ badges, active/inactive; tap → /shields.

**The 7-state render matrix** (fill from [COLOR §3.1]; the ConnectButton is the solid-fill+label pattern [COLOR §3.3]):

| ffi::TunnelStatus | Button fill<br>(light / dark)                              | Icon             | Label                             | StatusChip                          | Motion           |
|-------------------|------------------------------------------------------------|------------------|-----------------------------------|-------------------------------------|------------------|
| Disconnected      | grey<br>#6B7385 /<br>brand idle                            | power/<br>off    | <b>Not connected</b>              | "Tap to connect"                    | none             |
| Connecting        | amber<br>#B45309 /<br>#FBBF24                              | spinner-<br>ring | <b>Connecting...</b>              | "Contacting<br>exit..."             | connect          |
| Handshaking       | amber (= Connecting)                                       | spinner-<br>ring | <b>Securing...</b>                | "Handshaking..."                    | connect          |
| Connected{direct} | green<br>#15803D /<br>#22C55E                              | shield-<br>check | <b>Protected</b>                  | transport ·<br>direct · {rtt}<br>ms | settle<br>(shot) |
| Connected{relay}  | teal-green<br>#0F766E /<br>#2DD4BF                         | shield-<br>check | <b>Protected ·<br/>relayed</b>    | transport ·<br>relay · {rtt}<br>ms  | settle           |
| Reconnecting      | amber <b>pulse</b><br>#B45309 /<br>#FBBF24                 | rotating         | <b>Reconnecting...</b>            | "Lost link,<br>retrying..."         | connect          |
| Down{reason}      | orange<br>#C2410C /<br>#FB923C<br>(button) +<br>DownBanner | warning          | <b>Disconnected<br/>— dropped</b> | reason (stable-<br>prefix)          | none             |
| Danger{kind}      | red #DC2626 /<br>#F87171 + full<br>DangerBanner            | alert            | <b>Exposed / Kill-<br/>switch</b> | the danger<br>detail                | none (strobe)    |

**The loading-vs-connected distinction (§11.4.107).** A spinner is **not** "connected". Connecting/Handshaking/Reconnecting are **amber + animated + "..."-suffixed labels + a non-committal chip**; Connected is **green/teal + static settle + a shield-check icon + a chip showing real transport·path·rtt from the live Connected{...} payload**. The status is a **pure function of the FFI stream** [04\_CLIENT §8.2] — the button never shows "Protected" off a timer or an optimistic guess; only an actual Connected{...} event paints green. (This is the UI honouring the FFI's own no-fabrication rule [FFI §3.5].) The three signals — colour, icon, text — are independent so colour-blind and screen-reader users also get the distinction (§13 ally).

### States (non-connection).

- *loading (first frame / stream not yet seeded)*: the button renders the last cached status if any, else Disconnected as the safe default (never a fake "connected"); a subtle skeleton on the StatusChip until the first event.
- *offline (no network at all)*: Disconnected with chip "No network"; connect is attemptable (the shim surfaces the failure honestly) [04\_CLIENT §8.4].
- *error*: surfaced as Down/Danger per the stream; transient action errors (e.g. start failed) show a SnackBar with the typed reason, button returns to Disconnected.

### Responsive.

- *phone (compact)*: single centred column, ConnectButton dominant (~40 % height), chip + exit card + shields stacked below.
- *tablet (medium)*: the ConnectButton stays centred but a **side panel** appears showing the ExitSummaryCard + Shields as a persistent column (master/detail: tap exit → detail in the side pane without leaving Home).
- *desktop (expanded)*: ConnectButton left-centre; a right pane shows live StatusChip detail (throughput/rtt sparkline UNVERIFIED — owned by a future live-stats spec), exit, shields; nothing overlaps.
- *TV*: ConnectButton centred and **default-focused**; D-pad: down → exit card, down → shields row; the chip is enlarged; the connect action is the OK/center remote button.

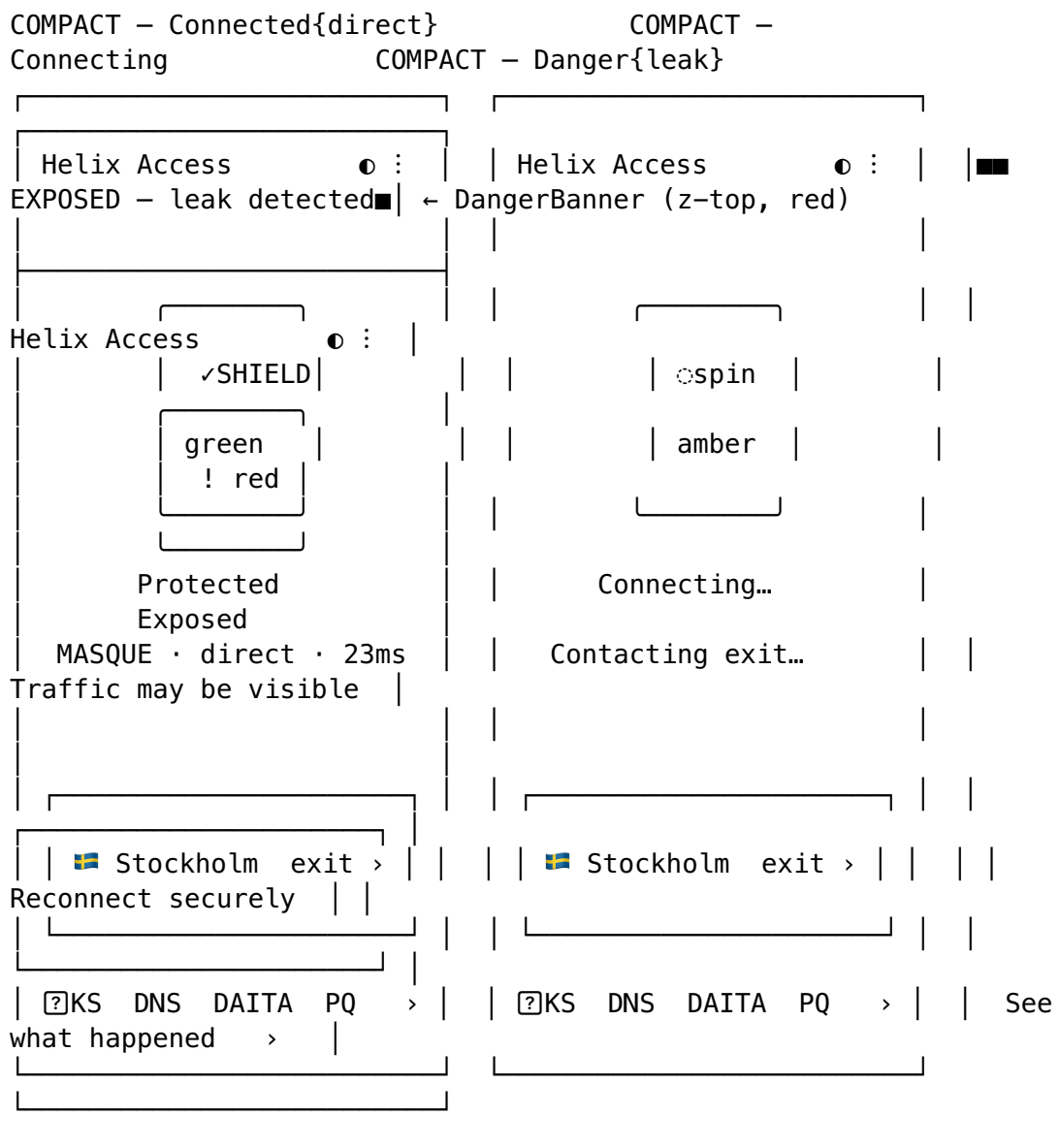
**Navigation.** in: post-onboarding root, or any nav destination's Home tab.  
out: ExitSummaryCard → /exits; Shields row → /shields; avatar → /account; Danger → pushes the danger surface (§11).

**Interactions.** tap **ConnectButton** → connectController.toggle(status) (connect if not Connected, disconnect if Connected) [04\_CLIENT §8.2]; tap exit → Exits; tap a shield badge → Shields (scrolled to that shield); long-press ConnectButton → quick transport override sheet (auto/specific) UNVERIFIED (transport-override UX pinned by [component-library.md]).

**A11y.** the ConnectButton has a **semantic label that announces the full state** ("Connected, direct path, 23 milliseconds" / "Connecting") and a role of toggle button; state changes fire a polite live-region announcement; the icon + label + colour triple means no colour-only signal (CI5 [04\_CLIENT §7.2]); large-text safe (the button scales, label wraps, never truncates under accessibility text sizes — no label overlay §11.4.162).

**Light + dark.** button fills per [COLOR §3.1] (both themes proven AA on the white / dark label, [COLOR §4.3]); the screen bg surface.default; the ExitSummaryCard

- shields on surface.raised; the connected "glow" ring uses a low-alpha tint of the state colour (UNVERIFIED exact glow alpha — U-COLOR-1 [COLOR §8]).



## 6. Exit selection

**Purpose.** Choose the exit gateway (or network) traffic leaves through: a searchable, RTT-sorted, favourite-able list with jurisdiction labels; and the entry point to the multi-hop builder (§7). Distinguishes a **privacy-exit** (a public/managed exit gateway) from a **network** (a private overlay network the device has joined) [04\_CLIENT §7.2 ExitPicker].

### Layout (regions).

1. SearchField (sticky top) — filters by name/city/country/jurisdiction.
2. SegmentTabs — **Exits** | **My networks** | **Favourites** (+ a "Multi-hop" entry).
3. The ExitPicker list of ExitRows, each: flag/glyph, name + city, jurisdiction chip (e.g. "🇨🇭 no-logs jurisdiction" — UNVERIFIED jurisdiction taxonomy, owned by the exits/catalog spec), **RTT** (live, colour-graded green→amber), a FavoriteToggle star, and a selected check.
4. A footer MultiHopEntry row → /exits/multihop.

### States.

- *loading*: skeleton rows while `core.exits()` resolves [04\_CLIENT §7.2].
- *empty*: "No exits available" (self-host with none configured) → CTA "Add a server" / "Join a network"; never a blank screen.
- *error*: "Couldn't load exits" + retry + the typed reason; the last cached list shown stale-labelled if available (honesty rule [04\_CLIENT §8.4]).
- *populated*: RTT-sorted ascending (fastest first); favourites pinned to top of their tab.
- *selected*: the active exit row carries a check + `surface.brandTint` background; selecting one updates Home's ExitSummaryCard and (if connected) triggers a seamless re-route (`apply_map/set_exit` [FFI §2]).

### Responsive.

- *phone*: full-width list, sticky search, bottom-nav present.
- *tablet (medium)*: master/detail — list left, selected-exit **detail pane** right (latency history, jurisdiction note, "make default").
- *desktop (expanded)*: same master/detail wider; multi-column row metadata.
- *TV*: large rows, D-pad vertical traversal, search via on-screen keyboard or voice (UNVERIFIED TV search input); favourite = remote long-press.

**Navigation.** in: Home ExitSummaryCard, or Exits nav destination. out: pick an exit → back to Home (state applied); "Multi-hop" → /exits/multihop.

**Interactions.** type to filter; tap a row to select; star to favourite; pull to refresh RTT; switch tabs (Exits/Networks/Favourites).

**A11y.** each row announces name + jurisdiction + RTT + selected-state + favourite state; RTT colour is paired with the numeric value (never colour-only); search field labelled; list is a single focus group with arrow traversal.

**Light + dark.** rows on surface.raised; selected row surface.brandTint; RTT colour grades green feedback.success → amber feedback.warning → (slow) red feedback.error, all AA both themes; jurisdiction chip neutral (surface.sunken + text.secondary).

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <div>  Search exits...         </div> <div> <span>[ Exits ]</span> <span>[ Networks ]</span> <span>[ ★ Fav ]</span> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <div>← SearchField (sticky)</div> <div>← SegmentTabs</div>                              |
| <div>  Stockholm           no-logs           18ms           ★✓         </div> <div>  Zürich           no-logs           24ms           ☆         </div> <div>  Frankfurt           EU           31ms           ☆         </div> <div>  New York           US-5eyes           88ms           ☆         </div> <div>... (RTT-sorted ascending)</div> | <div>← ExitRow (selected: brandTint + ✓)</div> <div>← RTT colour-graded + numeric</div> |
| <div>  Build a multi-hop chain           &gt;         </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <div>← MultiHopEntry → §7</div>                                                         |

## 7. Multi-hop builder

**Purpose.** Compose an ordered chain of exits (entry → ... → final exit) so traffic is relayed through multiple jurisdictions [04\_CLIENT §7.2 ExitPicker "multi-hop chain"]. Build, validate, and apply the chain.

**Layout.** A vertical HopChainEditor: an ordered list of HopRows (Hop 1 (entry) → Hop 2 → ... → Hop N (exit)), each a chosen exit with a remove control; an AddHopButton at the tail; a sticky ChainSummaryBar at the bottom (total estimated added latency =  $\Sigma$  hop RTTs, hop count, "Apply chain").

**States.**

- *empty (default)*: a single entry hop (= the current exit); building begins by adding a second.
- *building*: rows being added/reordered; live recompute of summed latency.
- *invalid*: a hop reachability/conflict problem (duplicate hop, an exit that cannot relay, overlapping CIDR if a network hop) → the offending row flags + the Apply button disables with a typed reason; never silently accept (§11.4.6).
- *valid*: Apply enabled; applying calls apply\_map/set\_exit [FFI §2] and the chain becomes active.

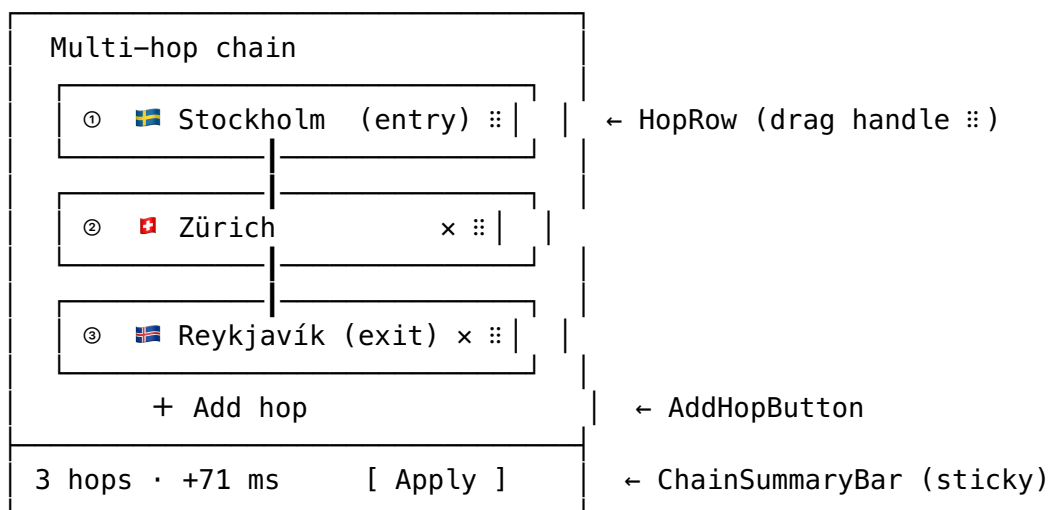
**Responsive.** phone: single column, drag-handle reorder; tablet/desktop: the chain on the left, a per-hop detail (latency, jurisdiction) on the right; TV: D-pad reorder via a "move mode", enlarged rows.

**Navigation.** in: Exits "Multi-hop" entry. out: Apply → Home (chain active) or Back (discard).

**Interactions.** add hop (opens an ExitPicker sub-sheet), reorder (drag / move mode), remove hop, apply.

**A11y.** each hop announces its ordinal + exit + jurisdiction; reorder is keyboard/remote operable (move-up/move-down actions), not drag-only; the summary bar announces total latency + validity.

**Light + dark.** hop rows `surface.raised` with an ordinal badge; the chain connector lines use `border.strong`; an invalid hop flags with `feedback.error` icon + text; summary bar `surface.raised` elevated (`elevation.semantic.menu`).



## 8. Shields / Privacy settings

**Purpose.** Control the privacy "shields" the tunnel enforces and view their honest cost: **kill-switch**, **DNS protection**, **DAITA** (defence-against-AI traffic-analysis), **post-quantum** (PQ key exchange), and **split-tunnel** (per-app/per-route bypass) [04\_CLIENT §6 splitTunnel, §7.2 ShieldIndicator "honest cost note on tap"]. Backed by the FFI `set_shields` call [FFI §2].

**Layout (regions).** A list of `ShieldIndicator` tiles, each a `SwitchTile`: icon + name + 1-line description + a toggle + an info affordance opening a `CostNoteSheet` (the honest trade-off — e.g. "PQ adds ~N ms handshake",

"DAITA adds constant-rate cover traffic, more battery/data"). Split-tunnel is a tile that expands into a SplitTunnelList (apps/routes to include or bypass).

### Per-shield state.

- *on*: toggle on, tile accent = `feedback.success` check; active badge mirrored on Home's shields row.
- *off*: toggle off, neutral.
- *unavailable*: the platform/exit can't offer it (e.g. DAITA not supported by the chosen exit) → toggle disabled + a "why" note; never a toggle that silently does nothing (§11.4.6).
- *pending*: applying a change is in flight (`set_shields` round-trip) → a small inline spinner on that tile; the rest stay interactive.

**Split-tunnel sub-list states.** loading apps / empty (all routed) / populated (per-app include-or-bypass toggles, searchable on platforms that enumerate apps — UNVERIFIED which platforms expose an app list; owned by the per-platform shim).

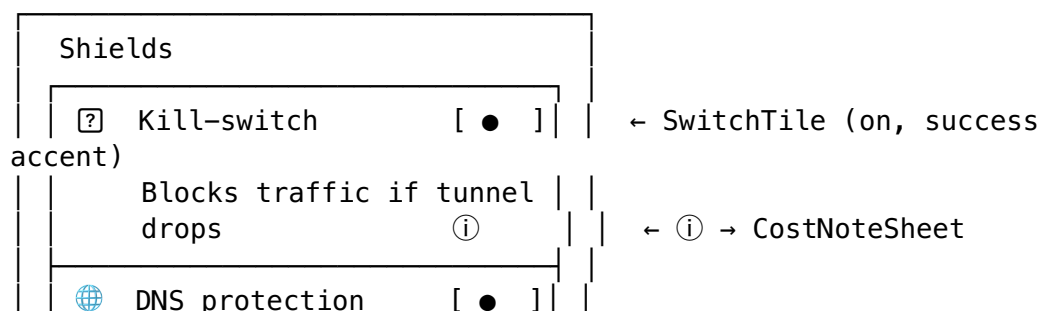
**Responsive.** phone: single column of tiles; tablet/desktop: two-column tile grid or master/detail (tile list | the selected shield's cost detail + split-tunnel editor); TV: enlarged tiles, D-pad traversal, cost sheets as focusable panels.

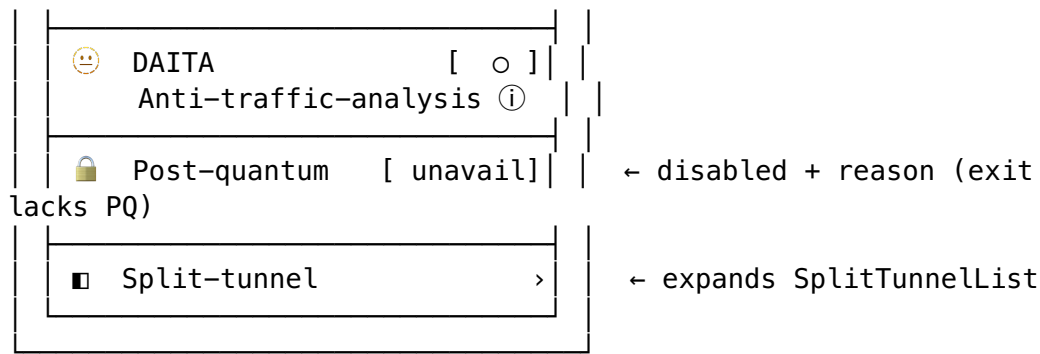
**Navigation.** in: Home shields row, or Shields nav destination. out: back to Home (changes apply live); split-tunnel deep into its editor.

**Interactions.** toggle a shield (applies via `set_shields`); open cost note; edit split-tunnel include/bypass lists; the kill-switch toggle carries an extra confirmation if turning it **off** (lowering protection).

**A11y.** each tile announces name + on/off + availability; the cost sheet is a labelled dialog; toggles are standard switch semantics; disabled (unavailable) toggles announce the reason; lowering protection (kill-switch off, split-tunnel bypass) is announced as a protection-reducing action.

**Light + dark.** tiles surface. `raised`; an **on** tile shows a `feedback.success` accent; an **unavailable** tile dims to `text.disabled` (exempt contrast, SC 1.4.3) with the reason in `text.secondary`; the kill-switch-off confirmation uses `feedback.warning`.





## 9. Account

**Purpose.** Show the device's identity/enrolment, the (optional) subscription, and the **device list** (other devices on this identity), with sign-out / unenroll. The Client's account capability [04\_CLIENT §6].

**Layout (regions).**

1. IdentityCard — who/what this device is enrolled as: OIDC identity (name/email
  - issuer) **or** the anonymous device-token label ("This device · anonymous"), plus the server it's bound to.
2. SubscriptionCard\* — plan/tier + renewal, if the deployment has billing.
3. DeviceList of DeviceRows — each other enrolled device (name, platform, last seen, this-device marker) with a per-row "remove device".
4. SignOutButton (and a destructive "Unenroll this device").

UNVERIFIED — **subscription/billing is not asserted for the self-hostable MVP**. The MVP's anonymous-device-token path implies **no billing** for self-host; SubscriptionCard renders **only** when the deployment advertises a billing capability. The plan/tier model, renewal copy, and purchase flow are owned by a future commercial-offering spec and are flagged UNVERIFIED here — not designed as fact. The IdentityCard + DeviceList are the asserted MVP surfaces.

**States.**

- *loading*: skeleton identity + device rows while helix\_api resolves.
- *loaded*: full data.
- *offline-cached*: last-known identity + devices shown with a "Last updated ..." stale label (honesty rule [04\_CLIENT §8.4]); device removal disabled offline.
- *error*: "Couldn't reach your server" + retry; identity from cache if present.
- *auth-revoked*: if the session/identity has been revoked server-side, Account is the natural place it surfaces → routes to /danger/auth (§11.c).

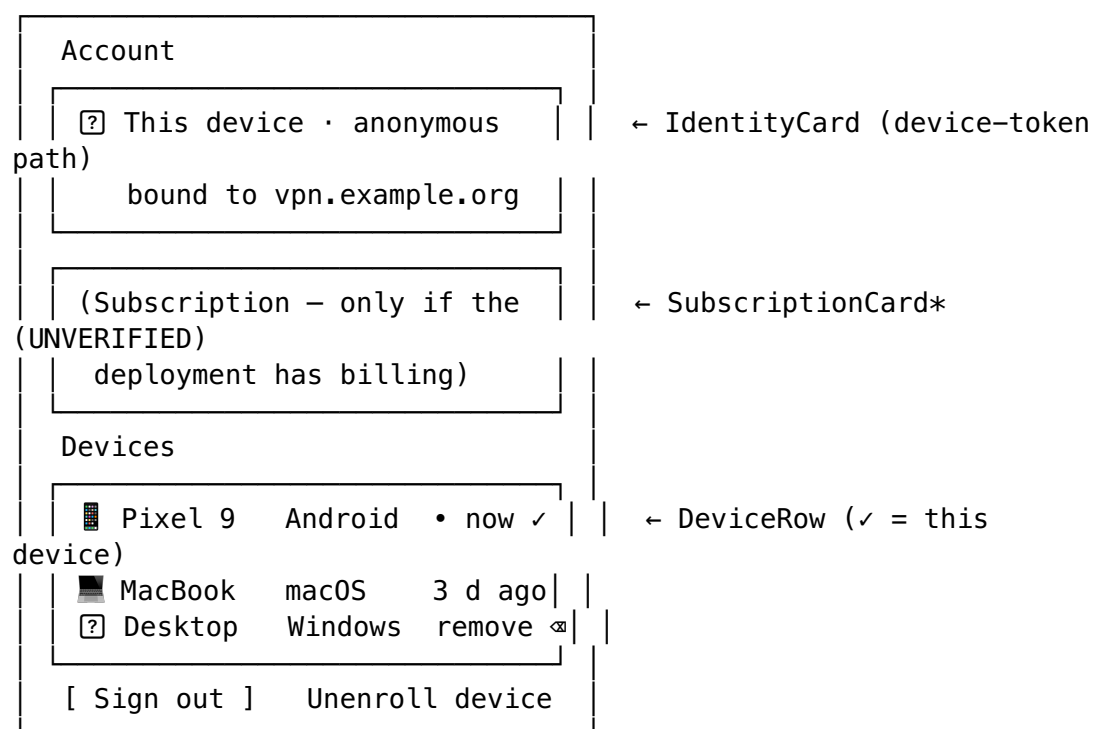
**Responsive.** phone: single column; tablet/desktop: identity + subscription in a left column, device list right (master/detail — tap a device → its detail); TV: enlarged cards, D-pad list traversal.

**Navigation.** in: Home avatar, or Account nav destination. out: sign-out → /welcome; auth-revoked → /danger/auth; device-detail (medium+) inline.

**Interactions.** remove another device (confirm); sign out (confirm); unenroll this device (destructive confirm — wipes the local token); copy the server URL.

**A11y.** identity announced incl. the privacy mode ("anonymous device"); each device row announces name + platform + last-seen + this-device; destructive actions carry confirming dialogs and are announced as destructive.

**Light + dark.** cards surface.raised; the current device marked with a feedback.info "this device" chip; destructive actions use feedback.error text on confirm; the anonymous badge uses a neutral chip (no identity leak in colour).



## 10. Settings

**Purpose.** General preferences, appearance, notifications, and about/diagnostics. Settings **recede** — the connection state owns the product; settings are a calm list [04\_CLIENT §7.2].

**Layout.** Grouped SettingsGroups:

- **General** — launch-on-startup, auto-connect (on trusted/untrusted networks), default transport (auto/specific), reconnect behaviour.
- **Appearance** → /settings/appearance (10.a).
- **Notifications** — connection events, drop/Danger alerts, updates (NotificationToggles).
- **Connector mode** → /settings/connector (§12), **only if** the connector capability is present (D-SC-3).
- **About** → /settings/about (10.b).

**States.** static list; each toggle is independently async (persists immediately, shows a momentary pending tick; failure reverts + a typed message — no silent drop §11.4.6).

**Responsive.** phone: single scrolling list; tablet/desktop: master/detail (group list left, the selected group's detail right — Settings is the canonical master/detail screen); TV: enlarged rows, D-pad.

**Navigation.** in: Settings nav destination. out: sub-pages (appearance, about, connector).

**A11y.** groups are landmark/heading nodes; toggles standard switch semantics; the selected sub-page is announced.

**Light + dark.** standard list on surface.default, rows surface.raised, section headers text.secondary.

|                      |          |                       |  |
|----------------------|----------|-----------------------|--|
| Settings             |          |                       |  |
| GENERAL              |          |                       |  |
| Launch on startup    | [ ● ]    |                       |  |
| Auto-connect         | >        |                       |  |
| Default transport    | Auto >   |                       |  |
| APPEARANCE           |          |                       |  |
| Theme                | System > | → 10.a                |  |
| NOTIFICATIONS        |          |                       |  |
| Connection events    | [ ● ]    |                       |  |
| Drop / Danger alerts | [ ● ]    |                       |  |
| ADVANCED             |          |                       |  |
| Connector mode       | >        | → §12 (if capability) |  |
| About                | >        | → 10.b                |  |

## 10.a Appearance (light / dark / system)

**Purpose.** Choose the theme — **Light**, **Dark**, or **System** (follow OS) — and preview the accent. Light + dark are both first-class (privacy users skew dark) [04\_CLIENT §7.1]; this screen is where the §11.4.162 light+dark mandate is user-visible.

**Layout.** A ThemeSelector (3-segment or 3 radio cards: Light / Dark / System)

- an AccentPreview showing a mini ConnectButton + StatusChip in the chosen theme so the change is visible before leaving.

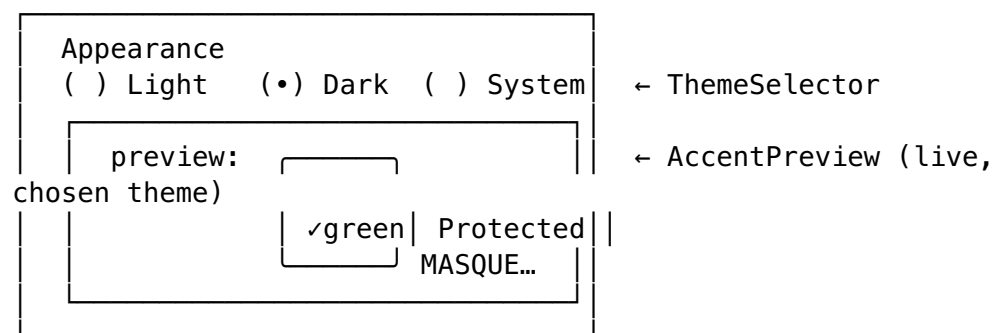
**States.** applied **immediately** on selection (the whole app re-themes live via the two ThemeData [DT §5.1]); "System" tracks OS changes live.

**Responsive.** phone: stacked cards + preview below; tablet/desktop: selector left, live preview right; TV: enlarged segments, default focus on the current choice.

**Navigation.** in: Settings → Appearance. out: back (choice persisted).

**A11y.** the three options are a labelled radio group announcing the current selection; the live preview is decorative-but-described ("preview of the connect button in dark theme").

**Light + dark.** this screen itself respects the chosen theme instantly; the preview demonstrates both connection-state palettes (proven [COLOR §3]).



## 10.b About

**Purpose.** Version, licenses, and a privacy-respecting diagnostics export.

**Layout.** AboutPanel — app name/version/build, server it's bound to, the Rust core version; LicensesList (open-source notices); a DiagnosticsButton (generates a redacted, user-reviewable diagnostics bundle — no traffic, no keys, per §11.4.10) + links (privacy policy, source).

**States.** static; diagnostics export shows a progress + a share/save result.

**Responsive.** phone: single column; desktop: two-column (info left, licenses right); TV: enlarged, focusable links.

**Navigation.** in: Settings → About. out: licenses detail; OS share sheet for diagnostics.

**A11y.** version is a readable row; licenses list is navigable; the diagnostics action announces what it includes/excludes (privacy-honest).

**Light + dark.** standard panel; version/build in text .secondary.

---

## 11. Error / Danger states

Two visual classes, by severity — this is the screen group that makes "am I exposed?" impossible to miss. **Down** (an unexpected drop, retrying/failed) is an **inline orange banner** on Home; **Danger** (leak / kill-switch tripped / auth revoked) is a **full red surface that overrides intent** (z-top, [FFI §3.2]).

### 11.a Leak detected — `Danger{kind:"leak"}`

**Purpose.** Tell the user, unambiguously and immediately, that traffic may be escaping the tunnel, and give the one safe action. Driven by the FFI projector's `shields_tripped` → Danger override [FFI §3.3], which **paints red regardless of where the user is**.

**Layout.** Full-screen DangerScreen: a z-top DangerBanner ("EXPOSED — leak detected"), a LeakDetail (what leaked — DNS/IPv6/route, in plain language, with the captured signal if available), and a single primary ReconnectButton ("Reconnect securely") + a secondary "See what happened" (the detail). Red #DC2626/#F87171 fill, white label, never occluded.

**States.** `Danger{kind:"leak"}` is the entry; on a successful re-establish the status stream returns to Connected and the surface dismisses to Home; if the kill-switch is on, traffic is already blocked (the screen says so — "Your traffic is blocked while exposed").

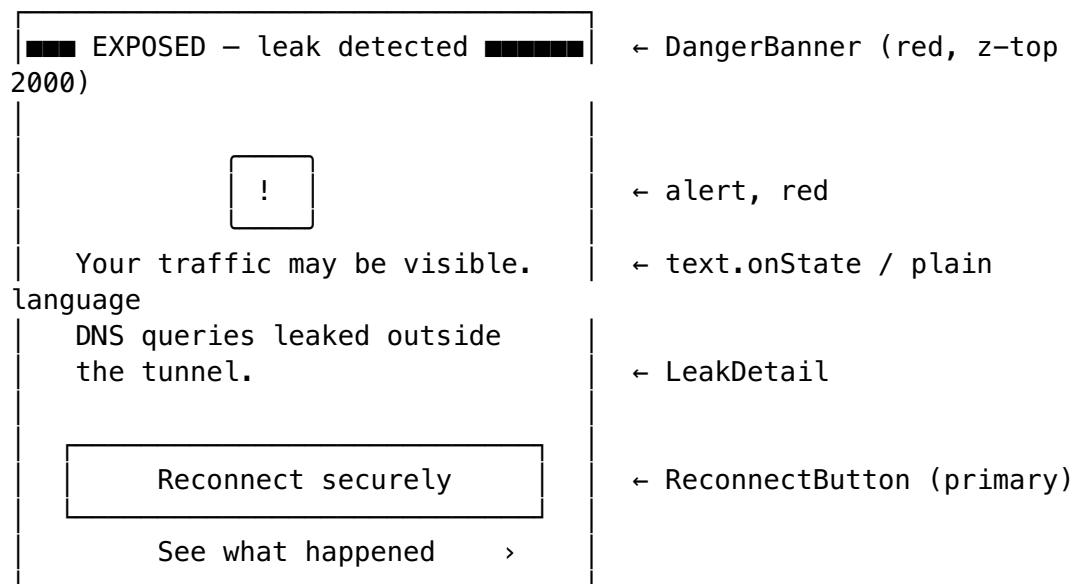
**Responsive.** all classes: centred, single decisive action; the banner spans full width; TV: the ReconnectButton is default-focused.

**Navigation.** in: pushed by any `Danger{leak}` event (overrides). out: reconnect ok → Home; "See what happened" → expands inline.

**Interactions.** Reconnect (re-runs connect); view detail; the screen cannot be swiped away while the danger persists (it is the truth, not a dismissible toast).

**A11y.** an **assertive** live-region announces "Exposed — leak detected" the moment it appears; the icon + "Exposed" text + red mean no colour-only signal; the single action is auto-focused.

**Light + dark.** red feedback.error/state.danger fill is identical-intent in both themes (safety colours never change [COLOR §6 D-COLOR-1]); the surface uses the danger palette over a scrim so nothing bleeds through.



## 11.b Kill-switch tripped — `Danger{kind:"killswitch_tripped"}`

**Purpose.** Explain that the kill-switch has **blocked all traffic** because the tunnel failed/auth-failed (the projector maps auth-failed → Danger [FFI §3.3]), and offer reconnect / sign-in-again / (last resort) disable kill-switch.

**Layout.** Full red DangerScreen + KillSwitchDetail ("All traffic is blocked to keep you private") + primary "Reconnect" + secondary "Sign in again" (if auth)

- a guarded "Turn off kill-switch" (protection-reducing, extra confirm).

**States.** `Danger{killswitch_tripped}`; resolves to Connected on a successful reconnect/re-auth, or the user explicitly lowers protection (confirmed).

**Responsive / a11y / light+dark.** as 11.a (assertive announce, default-focused primary, red both themes, guarded destructive action announced as protection-reducing).



reducing

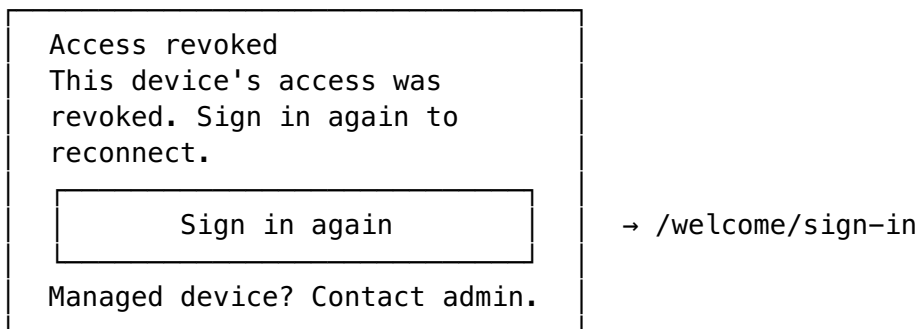
## 11.c Auth revoked

**Purpose.** The session/identity has been revoked server-side (admin removed the device, token expired/rotated). The device can no longer connect; it must re-enrol.

**Layout.** AuthRevokedScreen: explain ("This device's access was revoked"), single primary ReSignInButton → /welcome/sign-in (re-runs OIDC or device-token), and a "Contact your admin" note for managed deployments.

**States.** entered from Account / a 401-class API response / a Danger mapped from auth failure; exits via successful re-enrol → Home.

**Responsive / a11y / light+dark.** centred single action; assertive announce; uses feedback.warning/feedback.error framing (recoverable via re-sign-in, so warning unless traffic is exposed).



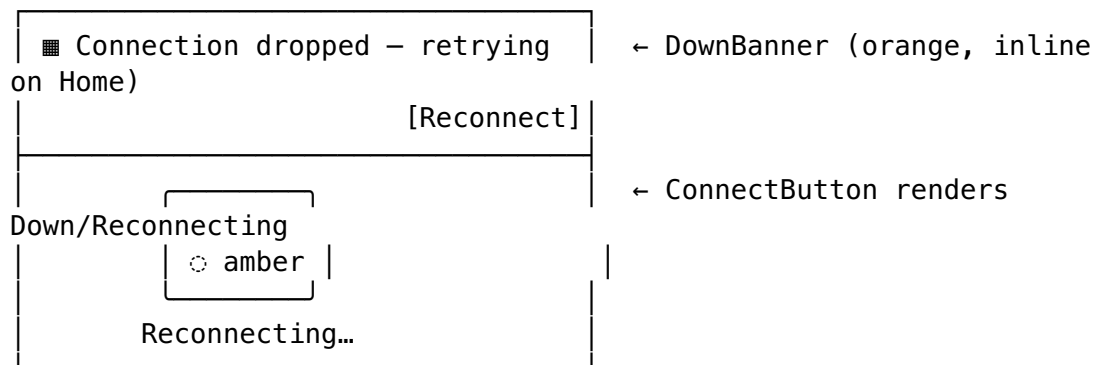
## 11.d Connection dropped — Down{reason} (inline banner)

**Purpose.** Distinguish an **unexpected drop that is being retried** / **has failed** (Down, orange) from an **exposure** (Danger, red). A drop is serious but not a leak — it is an **inline DownBanner on Home**, not a full red takeover ([FFI §3.2] splits Down from Danger; [COLOR §3.2] orange vs red).

**Layout.** A DownBanner strip at the top of Home (under the AppBar, above the ConnectButton): orange #C2410C/#FB923C, the stable-prefix reason in plain language ("Connection dropped — retrying" / "Couldn't reconnect"), and a Reconnect action. The ConnectButton beneath also reflects the orange Down state.

**States.** Down{reason}; while Reconnecting it shows "retrying" (amber pulse on the button); a successful re-dial → Connected and the banner clears; an exhausted ladder stays "Couldn't reconnect" with a manual Reconnect.

**Responsive / a11y / light+dark.** the banner spans the content width all classes; a **polite** (not assertive — it's a drop, not an exposure) live-region announce; orange feedback.warning-adjacent (distinct hue from red so colour-blind users read two alarms via the paired icon, [COLOR §3.2]); never overlays the ConnectButton label (sits above it, §11.4.162).



## 12. Connector-mode toggle

**Purpose.** Optionally let the Client device act as a **connector** — advertising local CIDRs into the overlay network so other peers can reach this device's LAN — *if* the build/deployment enables it. The Connector flavor's advertise / localAcl capabilities live primarily in the dedicated Connector app [04\_CLIENT §6]; surfacing a *toggle* inside the Client is a **surfaced decision** (D-SC-3) — when present it reuses the Connector components, gated behind the capability so the Client binary stays lean when it's absent (tree-shaken [04\_CLIENT §6]).

**Layout.** A ConnectorModeCard (master switch "Act as a connector") that, when on, expands an AdvertiseCidrList (the local CIDRs this device advertises, each with an add/remove + an **overlapping-CIDR conflict** surface [04\_CLIENT §6 table]) and a local-ACL note (→ the full editor lives in the Connector app / [screens-connector.md]).

### States.

- *unavailable*: the capability isn't compiled in → the whole Settings entry is **absent** (not a disabled stub) — tree-shaken [04\_CLIENT §6].
- *off*: master switch off; no advertising.
- *advertising*: switch on, CIDRs listed, peers can reach them; a live "advertising N routes" status.
- *conflict*: an advertised CIDR overlaps another peer's → the offending row flags with the conflict detail; advertising that row is blocked until resolved (§11.4.6 — never silently advertise a conflicting route).

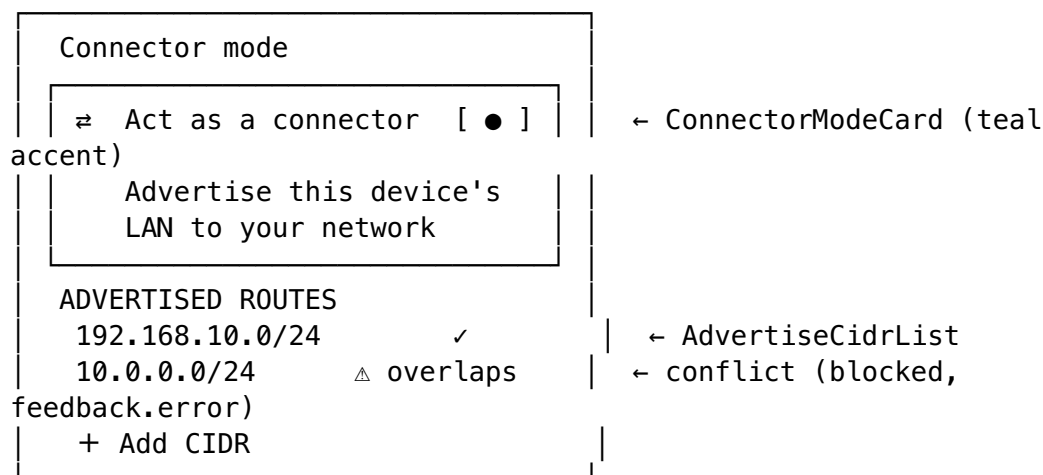
**Responsive.** phone: single column card + list; tablet/desktop: card + CIDR editor master/detail; TV: enlarged, D-pad (rare for a connector, but supported).

**Navigation.** in: Settings → Connector mode (capability-gated). out: deeper ACL config → the Connector surface [screens-connector.md].

**Interactions.** toggle connector mode; add/remove an advertised CIDR; resolve a conflict; (advanced) edit local ACL.

**A11y.** the master switch announces it changes this device's role (it now serves traffic, a meaningful posture change); each CIDR row announces its value + advertising state + any conflict; conflicts are announced.

**Light + dark.** the card uses the **Connector teal accent** [COLOR §6] when in the Client only for this surface's affordance (the connection-state safety colours are unchanged); a conflict row uses `feedback.error`.



## 13. Cross-screen patterns (loading/empty/error, a11y, motion)

These patterns apply to **every** screen so the spec above doesn't restate them.

### 13.1 The loading / empty / error triad

- **Loading** = a **skeleton** of the eventual layout (never a bare spinner on a blank screen), seeded from the **last cached value** where one exists so the user sees plausible structure immediately [04\_CLIENT §8.4]. The `ConnectButton`'s loading default is `Disconnected` — **never a fabricated Connected** (§11.4.107).

- **Empty** = an explanatory line + the single most useful CTA (add a server, join a network, add a CIDR) — never a blank pane.
- **Error** = a **typed, human message** + a retry, plus the stale-cached value shown stale-labelled when available (the honesty rule, [04\_CLIENT §8.4]). Errors are **never silent** (§11.4.6) and never a fake success.

## 13.2 Accessibility (every screen, CI5 [04\_CLIENT §7.2])

- **No colour-only signal.** Every connection state and every shield/danger state carries **colour + icon + text** (the load-bearing signal is the label, contrast- proven [COLOR §4]); ≈8 % of users have colour-vision deficiency ([COLOR §0]).
- **Screen-reader announcements.** State changes fire **polite** announcements; **Danger** fires **assertive**; the ConnectButton announces its full state.
- **Large text / high contrast.** Layouts reflow under accessibility text sizes with **no label overlap or truncation** (§11.4.162); text meets AA/AAA per [COLOR §4].
- **Keyboard / D-pad.** Full keyboard navigation on desktop; declared D-pad focus order on TV; every actionable has a visible border, focus ring (≥3.0 both themes [COLOR §4.5]); touch targets ≥48 dp (enlarged on TV) [04\_CLIENT §7.3].

## 13.3 Motion (reserved for state change [04\_CLIENT §7.2])

- The **connect transition** is the one place animation budget is spent: the ConnectButton morphs disconnected→connecting→connected, cross-fading **fill + label together** (motion.semantic.stateXfade 180 ms) so no in-between frame is illegible colour-on-colour (§11.4.162).
- Connecting/Reconnecting **pulse** (motion.semantic.connectPulse 1200 ms); under the OS **reduce-motion** flag the pulse degrades to **static amber** — it **never strobes** (a11y + §11.4.107 no-flash [DT §6.4]).
- Sheets/dialogs use motion.semantic.sheet (240 ms); presses motion.semantic.press (100 ms). All durations collapse to motion.semantic.reducedMotion (0 ms) under reduce-motion [DT §6.4].

## 13.4 Light + dark (every screen)

Every screen ships **both** themes (mandatory, §11.4.162 / gate CM-token-light-dark-complete [DT §9]); surfaces/text/borders from the semantic tokens [COLOR §2] (proven AA/AAA both themes), connection-state colours from [COLOR §3] (proven AA both themes), safety colours **identical** across themes and brands (D-COLOR-1 [COLOR §6]). The golden-screenshot/visual-regression suite renders **every screen × both themes ×**

**representative state** and asserts the [COLOR §4] contrast ratios + the no-overlap/no-label-overlay rule [visual-regression-and-qa.md], §11.4.162 / §11.4.168).

---

## 14. Surfaced decisions & cross-doc contracts

| id                   | Decision / contract                                                                                                                                                                                                                                                                                                                                                                | Status   |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| D-SC-1               | The Home ConnectButton renders <b>all 7</b> <code>ffi::TunnelStatus</code> variants with <b>colour + icon + text + an independent connected-payload check</b> — a spinner is never "connected" (§11.4.107).                                                                                                                                                                        | decided  |
| D-SC-2               | Down is an <b>inline orange banner on Home</b> ; Danger is a <b>full red, intent-overriding surface</b> (z-top). The split mirrors the FFI's Down vs Danger distinction [FFI §3.2].                                                                                                                                                                                                | decided  |
| D-SC-3<br>UNVERIFIED | Whether the <b>Client</b> exposes a connector-mode <b>toggle</b> (§12) vs requiring the separate <b>Connector app</b> is capability-gated and deployment-dependent; the toggle is specified here but its inclusion is a build decision (the Connector app remains the primary surface).                                                                                            | open     |
| D-SC-4<br>UNVERIFIED | <b>Subscription/billing</b> (§9 SubscriptionCard) is <b>not</b> asserted for the self-hostable MVP (anonymous device-token $\Rightarrow$ no billing); the card renders only when a billing capability is advertised. Plan/purchase UX is owned by a future commercial spec.                                                                                                        | open     |
| C-SC-A<br>(consumes) | The 7-variant <code>ffi::TunnelStatus</code> vocabulary + the <code>Connected{transport,path,rtt}</code> / <code>Down{reason}</code> / <code>Danger{kind}</code> payloads are owned by [FFI §3.2]; the Home matrix (§5) stays total over them. A new variant is a contract change here.                                                                                            | contract |
| C-SC-B<br>(consumes) | Connection-state colours + contrast proofs [COLOR §3–§4]; token scales (spacing/breakpoint/radius/motion) [DT §6]; the <code>AdaptiveScaffold</code> size-class contract [04_CLIENT §7.3]; the signature components ( <code>ConnectButton</code> / <code>StatusChip</code> / <code>ExitPicker</code> / <code>ShieldIndicator</code> / <code>NetworkTile</code> ) [04_CLIENT §7.2]. | contract |

| id                          | Decision / contract                                                                                                                                                                                                                                                                                                                                                           | Status   |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>C-SC-C</b><br>(provides) | This screen inventory + each screen's region/<br>state/responsive/a11y contract is the input the<br>component internals [component-library.md],<br>the routing config [navigation-and-routing.md],<br>and the visual-regression suite [visual-<br>regression-and-qa.md] build against.                                                                                        | contract |
| <b>U-SC-1</b><br>UNVERIFIED | Exact router paths/config (§1), TV per-screen<br>focus-order matrices (§0), the transport-override<br>sheet UX (§5), per-platform OS-consent copy +<br>privileged-service install (§4.3), jurisdiction<br>taxonomy (§6), and which platforms enumerate<br>apps for split-tunnel (§8) are pinned by their<br>named owner docs — stated as design intent<br>here, frozen there. | open     |

## Sources verified

- **The Client screen inventory, each screen's regions / responsive rules / state matrices / navigation / a11y / wireframes, the loading-vs-connected Home design, the Down-vs-Danger severity split, the onboarding three-step flow, the multi-hop builder, the shields layout, and all ASCII wireframes — NO external source needed — original HelixVPN UX design work** (owned by this document).
- **The 7-variant ffi::TunnelStatus vocabulary (Disconnected · Connecting · Handshaking · Connected{transport,path,rtt} · Reconnecting · Down{reason} · Danger{kind}), the Danger-overrides-intent rule, the Down-vs-Disconnected split, and the no-fabrication stream rule** — final/v04-client/ffi-surface.md §3.2–§3.5 (read 2026-06-25).
- **The Client flavor (access), its capability set {tunnel, account, splitTunnel}, the connector capabilities {advertise, localAcl}, the AdaptiveScaffold size-class contract (compact/medium/expanded), the signature components (ConnectButton/StatusChip/ExitPicker/ShieldIndicator/NetworkTile), the tunnel-state provider / toggle behaviour, the honesty rule for offline/cached/error, and the per-platform TunnelPlatform shim + OS-consent mechanisms (NEPacketTunnelProvider / VpnService / privileged service / Network Kit / tun)** — final/03-client-core-and-ui.md §4, §5, §6, §7, §8 (read 2026-06-25).
- **Connection-state palette, the connection-state colour mapping onto the 7 variants, WCAG contrast proofs, safety-colour stability (D-COLOR-1), per-app accents (the Connector teal used in §12),**

- and the no-overlap/no-label-overlay colour rules** — final/v10-design/color-system.md §3, §4, §5, §6 (read 2026-06-25).
- **Token scales referenced for spacing/breakpoints/radius/motion/z-index, the light+dark mandate + gate, and the reduce-motion contract** — final/v10-design/design-tokens.md §5, §6, §8, §9 (read 2026-06-25).
  - **3-app / 8-platform matrix, Web = Console-only (Client ships on the 7 native platforms), and the §11.4.162 OpenDesign no-overlap / light+dark / visual- regression mandate** — final/SPECIFICATION.md §3 + MASTER\_INDEX Volume 10 block (read 2026-06-25).
  - **Component internals (component-library.md), router config (navigation-and-routing.md), platform adaptation + TV focus order (platform-adaptation.md), Console/Connector screens (screens-console.md / screens-connector.md), and the visual-regression suite (visual-regression-and-qa.md)** — same-wave / planned sibling docs; forward-referenced as the owners of the detail this doc defers to. Items those docs own are tagged UNVERIFIED here per §11.4.6, not asserted as fact.
  - All UNVERIFIED items (D-SC-3 connector-in-client inclusion, D-SC-4 subscription/billing for self-host, U-SC-1 router paths / TV focus matrices / transport-override sheet / OS-consent copy / jurisdiction taxonomy / split-tunnel app enumeration) are pending their named owner docs — stated as design intent, never asserted as shipped fact.